Negative numbers, coordinates, and inequalities

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 10.2×10

2. $1000 \div 100$

3. Double 10.5

4. Half of 60

5. Find $|0|$ and describe it as a distance.

6. Find the opposite of 9.

7. A parallelogram has base 8, height 3, and slanted side 5 cm. Find its area.

8. A 10 by 4 cm rectangle loses a 3 by 1 cm corner. Find the remaining area.

9. Is $\frac{4}{3} \times 6$ smaller or larger than 6? Compute to check.

10. Find 25% of 24 stickers.

11. List common factors of 12 and 18. Find the greatest common factor.

12. List common factors of 8 and 20. Find the greatest common factor.

13. A point lies left of the y-axis and above the x-axis. It is 3 units from the y-axis and 2 from the x-axis. Find it, then reflect it across both axes in turn.

14. Write signed numbers for 7 meters below sea level and 2 meters above it. State what zero represents.

15. Order these values from least to greatest: $\frac{3}{2}$, -7, 0, -1.

16. Find the opposite of -7, then its opposite. Explain why you return to the start.

17. Find $|-7|$, $|7|$, and $|0|$. Compare the first two.

18. Which is warmer, $-7°C$ or $-9°C$? Which has greater absolute value?

19. Find the distance between -7 and 2 on a number line. Split the distance at zero.

20. Name the quadrant of each point: A(2, 7), B(−2, 7), C(−2, −7), D(2, −7).

21. Reflect (−2, 7) across the x-axis and then, starting again, across the y-axis.

22. Find the distance from (−2, −2) to (7, −2). Explain why only x changes.

23. A rectangle has vertices (−2, 0), (7, 0), (7, 3), (−2, 3). Find perimeter.

CHALLENGE / Explain your plan and check the result

C1. Points A and B lie on a horizontal line at $y = -2$. A is (−3, −2), and B is 8 units from A. Give both possible B points. Which is in quadrant IV? Reflect that point across the y-axis.

C2. Two shuttle lights flash every 12 seconds and 18 seconds. They flash together at time 0. Find the next shared flash and the number of flashes each light makes after time 0 up to and including that moment. Explain why GCF is not the time you need.